Experiment No.1: Generation and Classification of Continuous-Time and Discrete- Time Signals

OBJECTIVE:

(i) To generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal, and exponential signals.

(ii) To generate and visualize basic discrete-time signals and analyze their amplitude and time index characteristics.

(i) To generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal, and exponential signals.

Expected Outcomes

- Generate impulse, step, ramp, exponential and sinusoidal continuous-time signals.
- Obtain correctly scaled time-domain plots.
- Determine measurable parameters with preferably ≤5% deviation where applicable.
- Explain the generated waveform using its mathematical expression.

Software / Laboratory Environment :

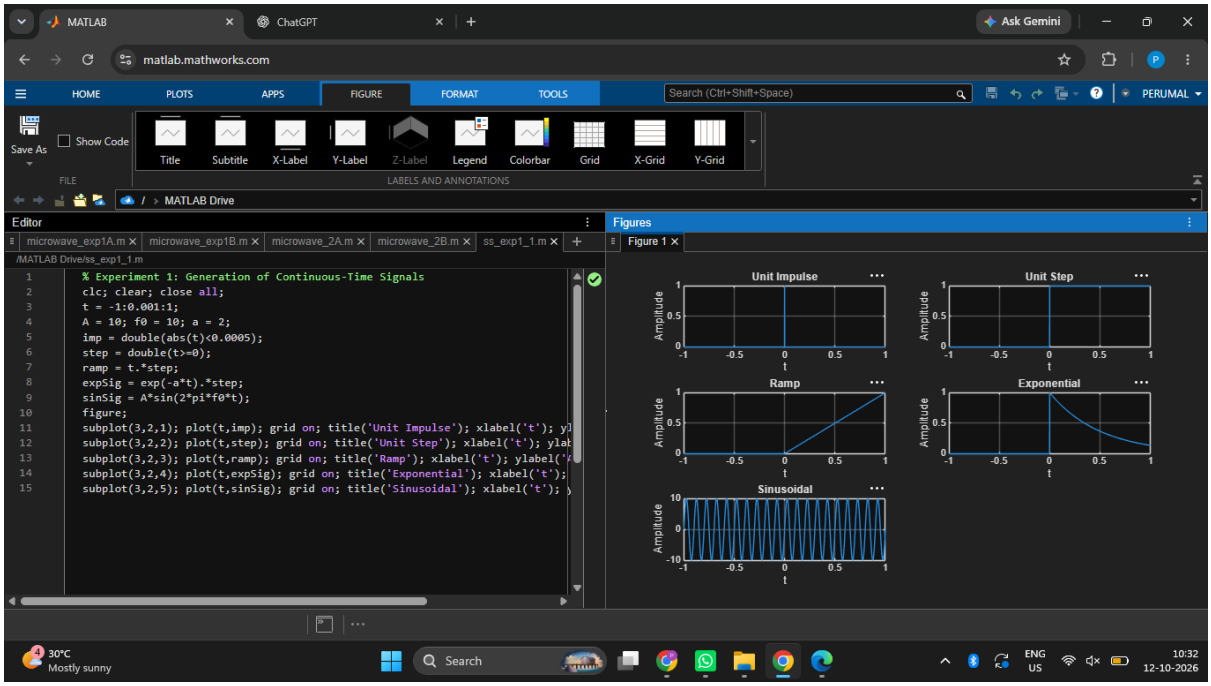
MATLAB

MATLAB Coding:

```
clc; clear; close all;
t = -1:0.001:1;
A = 2; f0 = 5; a = 1;
imp = double(abs(t)<0.0005);
step = double(t>=0);
ramp = t.*step;
expSig = exp(-a*t).*step;
sinSig = A*sin(2*pi*f0*t);
figure;
subplot(3,2,1); plot(t,imp); grid on; title('Unit Impulse'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,2); plot(t,step); grid on; title('Unit Step'); xlabel('t'); ylabel('Amplitude');
```

subplot(3,2,3); plot(t,ramp); grid on; title('Ramp'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,4); plot(t,expSig); grid on; title('Exponential'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,5); plot(t,sinSig); grid on; title('Sinusoidal'); xlabel('t'); ylabel('Amplitude');

OUTPUT:



RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks (10)	Marks obtained
1. Generate and visualize basic continuous-time signals such as unit step, unit impulse, ramp, sinusoidal and exponential.	**CT Signal Generation & MATLAB Implementation**	Generates all five CT signals accurately with a suitable time vector and step size. Uses correct MATLAB commands. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required CT signals and plots correctly, with minor errors in code, time range, parameters or labelling.	Generates basic CT signals but has noticeable errors in code, step or impulse representation, or plots, and needs assistance.	Unable to generate the required CT signals or produce meaningful plots.	**25**	**2.5**	

(ii) To generate and visualize basic discrete-time signals and analyze their amplitude and time index characteristics.

Expected Outcomes:

• Generate correctly indexed discrete-time sequences.

• Plot sequences using stem representation.

• Identify discrete-time waveform characteristics and periodicity.

• Validate the generated samples against theoretical expressions.

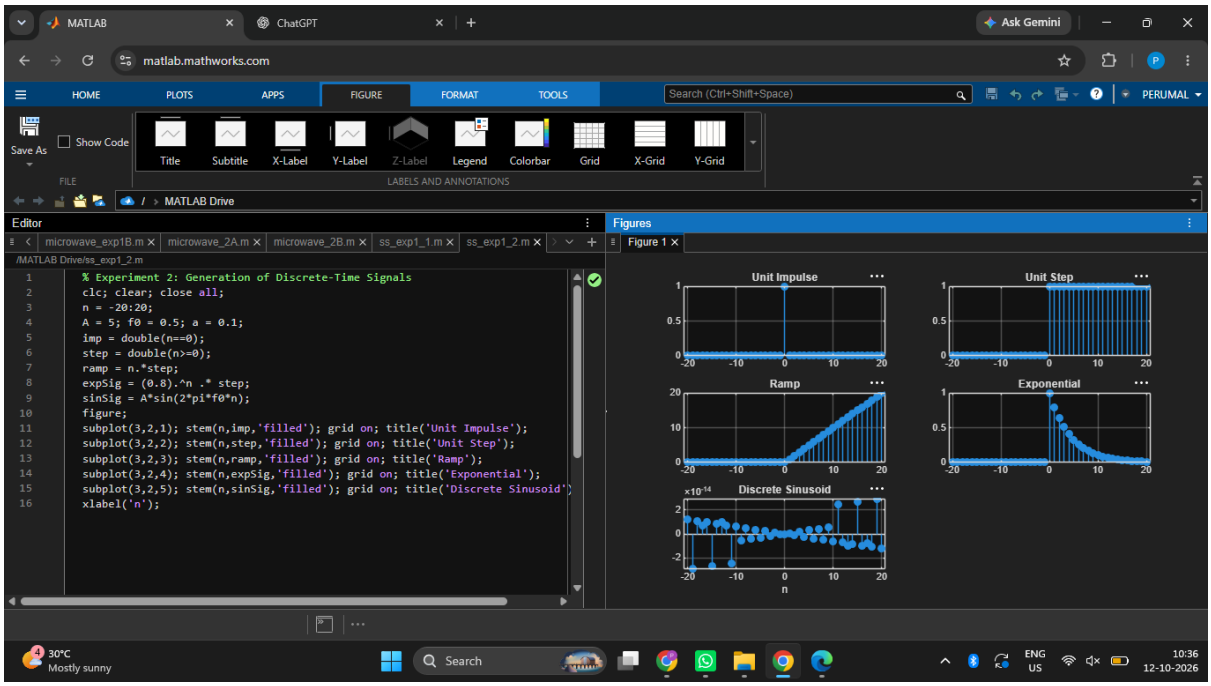
Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
clc; clear; close all;
n = -20:20;
A = 2; f0 = 0.1; a = 0.1;
imp = double(n==0);
step = double(n>=0);
ramp = n.*step;
expSig = (0.8).^n .* step;
sinSig = A*sin(2*pi*f0*n);
figure;
subplot(3,2,1); stem(n,imp,'filled'); grid on; title('Unit Impulse');
subplot(3,2,2); stem(n,step,'filled'); grid on; title('Unit Step');
subplot(3,2,3); stem(n,ramp,'filled'); grid on; title('Ramp');
subplot(3,2,4); stem(n,expSig,'filled'); grid on; title('Exponential');
subplot(3,2,5); stem(n,sinSig,'filled'); grid on; title('Discrete Sinusoid');
xlabel('n');
```

OUTPUT:



RESULT :

The continuous-time and discrete-time signals were generated and analyzed successfully.

RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks (10)	Marks obtained
2. Generate and visualize basic discrete-time signals such as unit sample, unit step, ramp, sinusoidal and exponential sequences.	**DT Signal Generation & MATLAB Implementation**	Generates all five DT sequences accurately with a proper index vector n. Uses stem correctly. Plots are correctly scaled, with titles, axis labels and grid.	Generates the required DT sequences and plots correctly, with minor errors in index range, stem usage, parameters or labelling.	Generates basic DT sequences but has noticeable errors in code, indexing or stem plots, and needs assistance.	Unable to generate the required DT sequences or produce meaningful stem plots.	**25**	**2.5**	